

BOSCOMBE DOWN, United KingdomWMO: **037460**Lat: **51.162N**Lon: **1.755W**Elev: **407**StdP: **14.48**Time Zone: **0.00 (EUW)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.5	27.9	20.3	15.4	30.4	22.9	17.5	31.7	35.5	49.9	31.5	48.8	6.8	300	0.633

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.9	79.6	64.7	75.9	63.0	72.7	61.5	66.3	75.7	64.5	72.6	63.0	69.9	10.0	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.0	87.4	69.0	61.6	83.2	66.9	60.3	79.4	65.2	31.1	75.6	29.7	72.5	28.6	70.0	73.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.6	24.3	21.4	DB	20.8	84.8	3.2	3.5	18.5	87.3	16.6	89.4	14.8	91.4	12.4	93.9	(4)
(5)			WB	20.2	69.5	3.2	2.0	17.9	70.9	16.0	72.1	14.2	73.2	11.9	74.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.5	40.3	40.9	43.9	47.9	53.4	58.5	62.4	61.9	57.7	52.1	45.3	41.1
	DBStd	9.49	5.85	5.51	5.14	5.18	5.17	4.49	4.53	4.28	4.06	4.93	5.48	6.49
	HDD50	1350	303	257	198	96	22	0	0	0	1	34	157	282
	HDD65	5351	767	675	655	512	360	201	108	116	223	401	591	742
	CDD50	1533	1	1	7	35	127	256	385	370	231	99	16	5
	CDD65	59	0	0	0	0	1	7	28	21	2	0	0	0
	CDH74	515	0	0	0	1	12	73	252	162	14	1	0	0
	CDH80	86	0	0	0	0	0	9	45	32	0	0	0	0
Wind	WSAvg	10.3	12.0	11.7	11.0	10.2	10.0	9.2	9.0	9.1	9.4	10.4	10.3	11.2
Precipitation	PrecAvg	28.1	3.0	2.0	2.2	1.9	2.1	2.1	1.6	2.2	2.6	2.6	2.8	3.1
	PrecMax	36.8	6.1	5.7	5.4	3.9	4.4	5.2	3.8	6.9	6.5	6.4	7.4	7.0
	PrecMin	19.5	0.3	0.1	0.1	0.1	0.4	0.2	0.3	0.2	0.3	0.2	1.0	0.6
	PrecStd	4.0	1.5	1.5	1.2	1.0	1.2	1.2	0.9	1.4	1.5	1.7	1.6	1.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.0	54.9	62.5	71.2	75.7	81.0	84.4	84.1	76.1	67.3	59.3	55.9
		MCWB	51.6	50.7	52.0	56.9	61.8	65.3	65.9	65.9	63.4	60.6	56.7	53.5
	2%	DB	52.6	52.3	57.5	64.4	71.5	75.6	80.4	78.6	71.6	63.3	57.1	54.3
		MCWB	50.5	49.3	49.7	53.6	59.4	62.5	65.2	64.7	62.0	57.9	54.9	52.4
	5%	DB	50.9	50.7	54.3	60.3	67.3	71.8	76.4	74.1	68.1	61.2	55.5	52.9
		MCWB	49.0	48.0	48.6	51.7	57.8	60.8	63.2	62.6	59.4	57.3	53.2	51.0
	10%	DB	49.2	49.1	51.9	57.0	63.2	68.4	72.7	70.6	65.5	59.5	53.9	51.0
		MCWB	47.3	46.6	47.8	49.8	55.2	59.0	61.7	61.0	58.2	56.3	51.7	49.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.5	52.1	53.2	58.5	63.8	67.4	68.7	68.8	65.7	61.9	57.6	54.3
		MCDB	53.4	54.1	58.1	67.9	71.7	77.3	80.1	79.8	72.0	65.6	58.6	55.1
	2%	WB	51.1	50.2	51.5	54.9	61.0	64.0	66.3	66.0	63.2	59.9	55.4	53.0
		MCDB	52.3	51.8	55.1	62.9	68.4	72.7	77.0	74.0	68.7	61.9	56.7	54.0
	5%	WB	49.2	48.6	50.1	52.7	58.6	62.0	64.4	64.2	61.5	58.4	53.8	51.3
		MCDB	50.6	50.1	53.2	58.3	65.5	69.4	73.4	71.0	65.7	60.3	55.2	52.6
	10%	WB	47.6	47.1	48.5	50.9	56.5	60.3	62.7	62.6	60.0	57.0	52.0	49.3
		MCDB	49.0	48.8	51.3	55.7	62.1	66.9	70.6	68.3	63.8	58.9	53.7	50.8
Mean Daily Temperature Range	5% DB	MDBR	9.8	10.8	13.2	15.5	15.9	16.4	16.9	16.0	15.1	12.3	10.7	9.5
		MCDBR	10.4	11.4	16.5	20.3	21.5	22.1	23.5	21.3	18.7	13.0	10.4	9.1
		MCWBR	10.0	9.2	10.5	11.7	12.0	11.6	10.6	9.6	9.9	8.9	8.9	8.8
	5% WB	MCDBR	10.4	10.7	13.9	17.7	19.1	19.4	20.3	17.6	15.9	11.3	10.0	9.4
		MCWBR	10.3	9.6	9.9	11.0	11.3	11.2	10.3	9.2	10.0	8.9	9.3	9.3
Clear-Sky Solar Irradiance	taub	0.324	0.339	0.375	0.396	0.399	0.394	0.389	0.380	0.367	0.351	0.325	0.313	
	taud	2.360	2.373	2.272	2.228	2.254	2.297	2.321	2.367	2.394	2.418	2.406	2.342	
	Ebn at Noon	218	248	259	267	271	273	273	270	260	240	216	201	
	Edh at Noon	20	26	34	40	41	40	38	35	30	24	19	18	
All-Sky Solar Radiation	RadAvg	259	477	841	1312	1590	1751	1657	1381	1035	591	312	203	
	RadStd	21	59	110	139	94	174	147	111	80	50	34	24	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47) Regional (73 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page